Paper II – Urology (MCh Urology, Winter 2017)

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II-1. Discuss principles of hemodialysis and vasular access in details.

Type: LAQ (25 marks)

Assumed meaning: "vasular" is a typo for "vascular" access for haemodialysis.

Introduction

- Haemodialysis (HD) removes solutes and water from blood across a semipermeable membrane in an extracorporeal circuit. It is the commonest KRT (kidney replacement therapy) for ESKD (end-stage kidney disease) and severe AKI (acute kidney injury). It needs an access giving about 250–350 mL/min of blood flow.

Principles of haemodialysis

Process	Driving force	What it removes
Diffusion	Concentration gradient across the membrane	Small solutes (urea, creatinine, potassium); bicarbonate enters blood
Convection	Transmembrane pressure; solute is dragged with water	Middle molecules (e.g. β2-microglobulin)
Ultrafiltration	Transmembrane (hydrostatic) pressure	Excess water, down to "dry weight"

- **Countercurrent flow:** dialysate runs opposite to blood, keeping a gradient along the whole dialyser. Clearance rises with flows, membrane area and time.
- **Dialyser:** hollow fibres of synthetic membrane; high-flux membranes clear more middle molecules.
- **Dialysate:** ultrapure (reverse-osmosis) water with sodium about 140 mmol/L, potassium about 2 mmol/L, bicarbonate and calcium, adjusted to the patient.
- **Anticoagulation:** heparin; heparin-free or regional citrate if bleeding risk is high (e.g. after surgery).
- **Schedule:** about 4 h three times a week. **Also know:** haemofiltration, haemodiafiltration (HDF), CRRT (continuous renal replacement therapy).

Adequacy

- **Kt/V** (dialyser urea clearance × time ÷ urea distribution volume): target 1.4, minimum 1.2 per thrice-weekly session (KDOQI [Kidney Disease Outcomes Quality Initiative] 2015). **URR (urea reduction ratio) ≥65%.**

Indications

- **Acute ("AEIOU"), refractory to medical treatment:** Acidosis; Electrolytes (hyperkalaemia); Ingestions (lithium, salicylates, alcohols); Overload with respiratory distress; Uraemia (encephalopathy, pericarditis, bleeding).
- **Chronic:** symptomatic ESKD, usually at eGFR (estimated glomerular filtration rate) 5–10; IDEAL (2010): early start gave no benefit.

Complications of haemodialysis

- **During dialysis:** hypotension (commonest), cramps, arrhythmia, air embolism, bleeding. **Disequilibrium syndrome:** rapid urea removal in early sessions shifts water into the brain (cerebral oedema); prevent with short, gentle first sessions.
- **Long term:** cardiovascular disease (main cause of death), mineral–bone disorder, calciphylaxis, β2-microglobulin amyloidosis (carpal tunnel), infection, hepatitis B and C; acquired cystic kidney disease (about 80% in ESKD; RCC [renal cell carcinoma] in 1–2% of these).

Types of vascular access

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino, 1966)	Radial artery to cephalic vein at the wrist (now usually vein end to artery side)	≥6–8 weeks	First choice: best patency, fewest infections; highest failure to mature
Brachiocephalic AVF	Brachial artery to cephalic vein at the elbow	≥6–8 weeks	Higher flow; more steal
Brachiobasilic AVF	Deep basilic vein transposed superficially	Longer	When the cephalic vein is unusable
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene), forearm loop or upper-arm straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; highest infection and central stenosis

- Non-tunnelled 10–12 Fr jugular or femoral catheters are for acute dialysis only.

Planning the access

- **Hierarchy:** AVF, then AVG, then CVC; distal before proximal; non-dominant arm first (US **Fistula First**, 2003). **KDOQI 2019** asks for an individual "ESKD Life-Plan": the right access for the right patient at the right time.
- **Timing:** refer at eGFR <30; create an AVF at least 6 months before dialysis is expected.
- **Vein preservation:** no venepuncture, cannula or PICC (peripherally inserted central catheter) in the non-dominant arm; avoid subclavian catheters.
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm), venography if central stenosis is suspected.
- **Maturity, "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, by 6 weeks.

Access complications

Complication	Key point	Treatment
Failure to mature	Juxta-anastomotic stenosis; side veins	Angioplasty; ligate branches

Complication	Key point	Treatment
Stenosis	Neointimal hyperplasia; AVG at venous anastomosis	PTA (percutaneous transluminal angioplasty), stent graft, revision
Thrombosis	Usually on an underlying stenosis	Thrombectomy, then treat the stenosis
Steal syndrome	Hand ischaemia; diabetes, brachial-based access	Banding, DRIL (distal revascularisation–interval ligation), ligation
Aneurysm	Repeated needling at one site	Repair if skin is thin
Infection	CVC > AVG > AVF	Antibiotics; excise graft; remove catheter

- **Also know:** venous hypertension (central stenosis); high-output cardiac failure (banding); ischaemic monomelic neuropathy; graft seroma.

Monitoring

- Each session: thrill and bruit, arm-elevation test (a healthy fistula collapses), prolonged bleeding after needling, rising venous pressure, falling Kt/V. Duplex scan when abnormal; treat a stenosis causing clinical change. Rotate needle sites ("rope-ladder").

Recent advances

- **Endovascular AVF** (percutaneous WavelinQ, Ellipsys); drug-coated balloons; home and nocturnal HD.
- **CONVINCE (2023):** high-dose HDF lowered all-cause mortality compared with high-flux HD.

Diagram to draw: dialyser with countercurrent blood and dialysate, and a radiocephalic end-to-side fistula.

> **Key points to remember:**
> - Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
> - Kt/V minimum 1.2 (target 1.4); URR ≥65%.
> - AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s.
> - Thrombosis usually sits on a stenosis; steal is treated by banding or DRIL.

II-2. Discuss evaluation for urolithiasis and management of bilateral renal staghorn calculi

Type: LAQ (25 marks)

Evaluation for urolithiasis

- **History:** colic, UTI (urinary tract infection), previous stones, fluid and diet, drugs (calcium, vitamins C and D, acetazolamide, steroids), GI (gastrointestinal) disease, gout, family history.
- **Laboratory:** urinalysis (pH, crystals, pyuria) and culture; serum creatinine, electrolytes, calcium, uric acid; PTH (parathyroid hormone) if calcium is high; bicarbonate if RTA (renal tubular acidosis) is suspected.

Imaging	Role
NCCT (non-contrast CT)	Gold standard: sensitivity 97%, specificity 95%. Size, site, density (uric acid 200–400 HU [Hounsfield units]; calcium oxalate 600–1200 HU), skin-to-stone distance
Low-dose NCCT (≤3 mSv)	Equal accuracy for stones ≥3 mm
Ultrasound	First line in pregnancy and children
KUB (plain X-ray)	Follow-up of radio-opaque stones only
CT urography	Anatomy before PCNL (percutaneous nephrolithotomy)
MAG3 (mercaptoacetyltriglycine) renogram or DMSA (dimercaptosuccinic acid) scan	Split function

- **Stone analysis:** infrared spectroscopy or X-ray diffraction.
- **Metabolic evaluation:** screening (history, serum, urine, stone analysis) for every first-time former. **High-risk formers** (recurrent, children, GI disease, obesity, diabetes, hyperparathyroidism, RTA, gout, family history, solitary kidney, large bilateral stones) need one or two **24-h urines** on a usual diet: volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine (cystine if suspected). Bilateral staghorns are high risk by definition.

Staghorn calculus

- **Definition:** a branched stone filling the renal pelvis and extending into the calyces. **Partial:** pelvis and some calyces; **complete:** pelvis and all major calyces.
- **Composition:** classically struvite (magnesium ammonium phosphate) with carbonate apatite. Bacterial **urease** (Proteus, Klebsiella, Ureaplasma) splits urea into ammonia, so urine pH rises above 7.2 and struvite crystallises inside a bacterial biofilm. Fewer than half of modern staghorns are infection stones; calcium phosphate, cystine and uric acid also occur.
- **Why treat:** untreated staghorns destroy the kidney and cause sepsis, XGP (xanthogranulomatous pyelonephritis) and death.

Work-up before surgery

- NCCT with coronal and 3D views: burden, calyceal anatomy, colon and spleen.
- **Renal function:** eGFR (estimated glomerular filtration rate) and split function, because both kidneys are at risk.
- **Infection:** culture-specific antibiotics before surgery; drain an obstructed or infected kidney (nephrostomy through a calyx usable later for PCNL). Culture the stone.
- **Guy's stone score (2011)** predicts stone-free rate:

Grade	Stone
I	Single stone in mid or lower pole or pelvis, simple anatomy
II	Single upper-pole stone; multiple stones in simple anatomy; or single stone in abnormal anatomy
III	Multiple stones in abnormal anatomy; calyceal diverticulum; partial staghorn
IV	Staghorn; or any stone in spina bifida or spinal injury

- **Also know:** STONE score, CROES nomogram, S-ReSC score.

Treatment options (AUA [American Urological Association], EAU [European Association of Urology])

- **Goal: complete clearance**, because each fragment is an infected nidus; also clear infection and preserve function.

Option	Role
PCNL	**First line.** Multiple tracts if needed; upper-pole access reaches pelvis, lower pole and ureteropelvic junction. Flexible nephroscopy raises clearance; second-look for residue
ECIRS (endoscopic combined intrarenal surgery)	PCNL (usually supine) plus retrograde flexible ureteroscopy: fewer tracts, less bleeding, higher clearance
Staged RIRS (retrograde intrarenal surgery)	Selected partial staghorns; patients who cannot stop anticoagulants
SWL (shock wave lithotripsy)	Not as monotherapy; adjunct for residual fragments
Anatrophic nephrolithotomy (Boyce–Smith)	Incision along Brödel's avascular line, artery clamped, cold ischaemia, infundibula reconstructed. Now rare; when endoscopy cannot clear the stone in a reasonable number of sessions
Nephrectomy	Poorly functioning kidney (split below about 10–20%) that is infected or painful; XGP

Bilateral staghorns: sequence

```
Sepsis or obstruction on either side?
  Yes -> drain + antibiotics -> treat that side once sterile
  No  -> treat the better-functioning (or symptomatic) kidney first
              |
  NCCT: stone-free? -> second look if residue
              |
  Other kidney after recovery
  (selected: same-session bilateral PCNL, or PCNL + contralateral RIRS)
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- **Why the better kidney first:** it secures renal function early; some start with the simpler side. No trial defines the order.
- **Same-session bilateral surgery:** feasible at experienced centres if the first side was short and uncomplicated; saves an anaesthetic but risks bilateral bleeding or obstruction.
- Check creatinine after each side. Remove a non-functioning kidney only after the other side is cleared.

Complications and follow-up

- PCNL risks: bleeding (transfusion about 1%; angioembolisation), sepsis, pleural, colon, spleen or liver injury.
- NCCT for stone-free status; treat residual fragments >4 mm. Surveillance cultures.
- 24-h urine even in struvite formers, then targeted therapy; urine ≥2.5 L/day.

- **AHA (acetohydroxamic acid)** 250 mg three times daily, a urease inhibitor, only when surgery is exhausted. Avoid if creatinine >2.5 mg/dL or in pregnancy; causes DVT (deep vein thrombosis), haemolytic anaemia.

Diagram to draw: staghorn with an upper-pole PCNL tract and a retrograde flexible scope (ECIRS).

> **Key points to remember:**
> - NCCT gold standard (97%/95%); 24-h urine for high-risk formers.
> - Staghorn: PCNL first line; complete clearance is the goal.
> - Bilateral: drain an infected side first, then the better kidney.
> - AHA only when surgery is not possible.

II-3a. LASERS in BPH.

Type: SAQ (10 marks)

Principle

- LASER (light amplification by stimulated emission of radiation): the tissue effect depends on wavelength and **chromophore** (water or haemoglobin), which set **penetration depth**.
- About 60 °C coagulates tissue (delayed sloughing); ≥100 °C vaporises it (immediate removal).

Lasers used in BPH (benign prostatic hyperplasia)

Laser	Wavelength	Chromophore	Penetration	Procedure
Ho:YAG (holmium:yttrium-aluminium-garnet)	~2120 nm	Water	0.4 mm	**HoLEP** (holmium laser enucleation of the prostate)
Thulium (Tm:YAG, TFL [thulium fibre laser])	2013 / 1940 nm	Water (TFL at peak)	0.2 mm	ThuLEP (thulium laser enucleation), vaporesection, vaporisation
KTP (potassium titanyl phosphate) / LBO (lithium triborate)	532 nm	Haemoglobin	0.8 mm	**PVP** (photoselective vaporisation), 120–180 W
Diode	940–1470 nm	Water and haemoglobin	0.5–5 mm	Vaporisation, enucleation
Nd:YAG	1064 nm	Weak	About 1 cm	Coagulation; abandoned (sloughing)

Enucleation versus vaporisation

Feature	Enucleation (HoLEP, ThuLEP)	Vaporisation (PVP)
Tissue	Whole transition zone to the capsule, then morcellated	Removed layer by layer
Histology	Yes	No
Gland size	**Any (size-independent)**	Up to 80 cc
Retreatment	Rare	Higher, especially >80 cc

Evidence

- **HoLEP** (about 2 J × 40–50 Hz): the only endoscopic procedure **superior to TURP** (transurethral resection of prostate) for IPSS (International Prostate Symptom Score) and Qmax. Less bleeding, shorter catheter time and stay.
- **GOLIATH RCT** (randomised controlled trial; 180 W PVP vs TURP, about 270 men, ≤100 cc): PVP non-inferior for IPSS, Qmax and complications at 2 years; shorter catheter time and stay.
- **ThuLEP:** outcomes equal to HoLEP. TFL energy is absorbed about four times more by water than holmium.
- **AUA (American Urological Association):** HoLEP or ThuLEP for any size; PVP for <80 cc; HoLEP, ThuLEP or PVP for men at **higher bleeding risk**.

Complications

- Storage symptoms and dysuria; transient stress incontinence; retrograde ejaculation (about three-quarters after HoLEP); bladder neck contracture (0–3%); stricture; capsular

perforation; morcellator bladder injury.

- **Safety:** wavelength-specific eyewear for all; laser safety officer.

Diagram to draw: HoLEP planes: enucleated adenoma pushed into the bladder, then morcellated.

> **Key points to remember:**
>
> - Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
> - HoLEP is size-independent and superior to TURP.
> - GOLIATH: PVP non-inferior to TURP; best for anticoagulated men.

II-3b. Interstitial cystitis (Bladder pain syndrome).

Type: SAQ (10 marks)

Definitions

- **AUA (American Urological Association):** an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the bladder, with LUTS (lower urinary tract symptoms) for **more than 6 weeks**, without infection or another identifiable cause.
- **ESSIC (European Society for the Study of Interstitial Cystitis, 2008):** chronic (>6 months) bladder-related pain with at least one other urinary symptom (persistent urge or frequency), after confusable diseases are excluded.
- NIDDK (National Institute of Diabetes and Digestive and Kidney Diseases, 1987) research criteria missed about 60% of patients.

Phenotypes

- **Hunner lesion (HL) disease, "IC":** about 20%, mainly over 50; true inflammation (lymphoplasmacytic infiltrate, mast cells).
- **Non-Hunner BPS (bladder pain syndrome):** bladder normal or shows glomerulations.
- **ESSIC type:** cystoscopy 1 normal, 2 glomerulations, 3 Hunner lesion; biopsy A normal, B inconclusive, C positive (e.g. 3C).

Pathogenesis (theories)

- Leaky GAG (glycosaminoglycan) layer; mast-cell activation; neurogenic inflammation and central sensitisation; stress-axis dysfunction; small-fibre neuropathy. No infection proven. 75–80% have pain beyond the pelvis (IBS [irritable bowel syndrome], fibromyalgia).

Evaluation (diagnosis of exclusion)

- History (**ask about ketamine**), pelvic-floor palpation, PVR (post-void residual), urinalysis, culture, voiding chart, symptom scores; cytology if risk factors.
- **Office cystoscopy if over 50 or HL suspected;** biopsy to exclude CIS (carcinoma in situ).
- **Local anaesthetic challenge** localises pain to the bladder. Potassium sensitivity test is not recommended.
- **Also know:** urodynamics optional; hydrodistension glomerulations are non-specific.

Treatment (AUA 2022: phenotype-based, no fixed tiers)

Group	Options
Behavioural	Education, dietary triggers, stress; **pelvic-floor myofascial therapy** if tender
Oral	**Amitriptyline** 10 mg rising to about 50 mg; hydroxyzine; cimetidine; **PPS** (pentosan polysulfate) 100 mg three times daily (**pigmentary maculopathy**: eye checks)
Intravesical	**DMSO** (dimethyl sulfoxide) 50%, 50 mL, 15–20 min dwell; heparin; alkalinised lidocaine
Procedures	**HL: fulguration and/or triamcinolone injection** (75–90% improve); short low-pressure hydrodistension; onabotulinumtoxinA; sacral neuromodulation
Last resort	Ciclosporin A for refractory HL; diversion ± cystectomy (<10%)

- **Do not offer:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term systemic steroids.

Diagram to draw: stellate Hunner lesion with radiating vessels.

Key points to remember:

- AUA definition: bladder pain with LUTS >6 weeks, no other cause.
- Hunner lesion: cystoscopy over 50; fulguration or triamcinolone.
- Amitriptyline, PPS, DMSO; potassium test obsolete.

II-3c. Silodosin.

Type: SAQ (10 marks)

Pharmacology

- The **most α1A-selective** blocker (well over 100-fold vs α1B; about 50-fold vs α1D), used for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).
- Sparing α1B gives **minimal hypotension**. Blocking α1A in the vas and seminal vesicles causes **failure of emission**.
- Onset: flow improves within 2–6 hours, symptoms within 3–4 days. Metabolised by CYP3A4.

Dose and contraindications

Item	Detail
Dose	**8 mg once daily with a meal**
CrCl (creatinine clearance) 30–50 mL/min	**4 mg daily**
Contraindicated	CrCl <30; severe liver impairment; strong CYP3A4 inhibitors (e.g. ketoconazole)
Caution	Other α-blockers, PDE5 (phosphodiesterase 5) inhibitors, planned cataract surgery

Efficacy

- Better than placebo for IPSS (International Prostate Symptom Score) and flow, including severe LUTS. **Non-inferior to tamsulosin 0.4 mg**, not better (European trial, 2011).
- **Greatest fall in BOOI (bladder outlet obstruction index)** of all α1-blockers.
- Does not shrink the prostate; add a 5-ARI (5α-reductase inhibitor) for large glands.

Adverse effects

- **Anejaculation:** over a quarter of men in trials (about 14% vs 2% with tamsulosin head-to-head), yet few stop. Men with anejaculation had better responses.
- No excess hypotension in the elderly or with antihypertensives; no QT prolongation.
- **IFIS (intraoperative floppy iris syndrome):** not specifically studied but reported; warn before cataract surgery.

Other uses

- **MET (medical expulsive therapy)** for distal ureteric stones 5–10 mm. α-blockers reduce ureteric spasm. Silodosin gave faster expulsion than tamsulosin in meta-analyses; **SUSPEND (2015)** found no overall tamsulosin benefit.
- **Also know:** before ureteroscopy (easier access); recent-onset CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome).

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction
Silodosin	α1A (highest)	8 mg	Lowest	**Highest**
Tamsulosin	α1A, α1D > α1B	0.4 mg	Low	Moderate
Alfuzosin	Clinically uroselective	10 mg	Low	Lowest
Doxazosin, terazosin	Non-selective; titrate	1–8 mg, 1–10 mg	Highest	Low

Diagram to draw: α1A in prostate and vas, α1B in vessels, α1D in bladder.

> **Key points to remember:**
>
> - 8 mg with a meal; 4 mg if CrCl 30–50; avoid if <30.
> - Equal to tamsulosin 0.4 mg; greatest BOOI fall.
> - Anejaculation is the signature effect; least hypotension.

II-3d. ADPKD (Adult polycystic kidney disease)

Type: SAQ (10 marks)

Genetics

- ADPKD (autosomal dominant polycystic kidney disease): commonest inherited cystic kidney disease (about 1 in 400–1000); 7–15% of patients on haemodialysis.

Gene	Chromosome	Share	ESRD (end-stage renal disease)
PKD1 (polycystin-1)	16	~85%	50s
PKD2 (polycystin-2)	4	~15%	70s

- Polycystins (primary cilium) restrain proliferation.

Clinical features (30–50 years)

- **Pain** (commonest): mass, cyst haemorrhage, infection, stones. **Haematuria** about 50%; **stones** 20–30%.
- **Hypertension:** about 50% at 20–35 years with normal function; stretched vessels release renin.
- **Renal failure:** tracks kidney volume; worse with PKD1, male sex.
- **Extrarenal:** liver cysts (commonest; up to 90% by 50); **intracranial aneurysm 10–30%**; seminal vesicle (40%), pancreatic, arachnoid cysts; mitral valve prolapse; diverticulosis.
- RCC (renal cell carcinoma) is not commoner, but is more often bilateral and sarcomatoid.

Diagnosis (ultrasound in at-risk relatives)

Age	Ravine (PKD1 families)	Pei unified (2009)
15–29	≥2 cysts, one or both kidneys	≥3 cysts in total (15–39)
30–59	≥2 cysts in each kidney	≥2 in each kidney (40–59)
≥60	≥4 cysts in each kidney	≥4 in each kidney

- CT or MRI shows TKV (total kidney volume), which predicts progression. Genetic testing if equivocal or for a living donor.

Management

- **BP <130/80 mmHg** with ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker); HALT-PKD: tighter control slowed TKV growth.
- **Tolvaptan** (V2-receptor antagonist; lowers cyclic adenosine monophosphate): **TEMPO 3:4 (2012)** slowed TKV growth and function loss; **REPRISE (2017)** slowed eGFR (estimated glomerular filtration rate) decline in later disease. Causes polyuria and liver toxicity.
- **Pain:** exclude infection, stone, tumour; avoid NSAIDs (non-steroidal anti-inflammatory drugs); aspiration, then laparoscopic unroofing.
- **Infected cyst:** lipid-soluble antibiotics (trimethoprim–sulfamethoxazole, fluoroquinolones); drain if no response.
- **Haemorrhage:** usually settles with rest. **Aneurysm screening** (MR angiography) if family history.
- **ESRD:** dialysis or transplant (no recurrence). **Native nephrectomy** before or at transplant for recurrent infection, bleeding, pain, suspected tumour or no room for the graft.

Diagram to draw: large cystic kidneys, liver cysts and berry aneurysm.

Key points to remember:

- PKD1: chromosome 16, 85%, earlier ESRD.
- Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
- Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

II-3e. RIRS (Retrograde intrarenal surgery)

Type: SAQ (10 marks)

Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureterorenoscopy (fURS) inside the kidney through the natural route; active deflection and the holmium laser made it practical.

Instrument

Feature	Details
Tip	4.5–7.5 Fr, tapered (digital tips up to about 8.5 Fr)
Working channel	3.6 Fr (laser fibre, basket)
Deflection	≥270° both ways; lower pole needs about 140° (up to 175°)
Types	**Fibreoptic** (smallest, image degrades); **digital** (chip on tip, better image, not more durable); **single-use** (7.7 Fr; cheaper below about 99 cases a year)

Indications

- **Stones:** renal stones <2 cm; lower pole <1 cm (or >1 cm as an alternative to PCNL [percutaneous nephrolithotomy]); patients who cannot stop anticoagulants; failed SWL (shock wave lithotripsy); obesity; staged treatment of larger stones.
- **Others:** diagnosis and ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects; endopyelotomy; calyceal diverticulum; essential haematuria.

Technique

1. Sterile urine; single-dose antibiotic.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.
3. **Access sheath** (usually 12/14 Fr) over a stiff wire: lowers intrarenal pressure and allows repeated passes, but severe ureteric injury occurs in about 1 in 8.
4. Low-pressure irrigation. **Ho:YAG (holmium:yttrium-aluminium-garnet) or thulium fibre laser**: dusting (low energy, high frequency) or fragment-and-basket. Keep power ≤10 W in the ureter and ≤25 W in the kidney (tissue damage above 43 °C).
5. Inspect the ureter on withdrawal; stent unless uncomplicated.

- **Difficult lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum. Relocate the stone to an upper calyx with a basket.

Complications

- Sepsis about 5%; stricture about 3%; bleeding or subcapsular haematoma <1%; perforation; avulsion <1%; stent symptoms; scope damage.

Recent advances

- Single-use digital scopes; **thulium fibre laser** and pulse-modulated (Moses) holmium (less retropulsion, finer dust); **suction and flexible navigable sheaths** (FANS); intrarenal pressure sensing; robotic flexible ureteroscopy.

Diagram to draw: flexible scope deflected into the lower pole calyx through an access sheath.

> **Key points to remember:**
> - 3.6 Fr channel; ≥270° deflection.
> - Access sheath lowers pressure but injures the ureter in about 1 in 8.
> - Laser power ≤10 W ureter, ≤25 W kidney.

II-3f. Renal isotope scans

Type: SAQ (10 marks)

Principle

- A radiotracer follows a physiological process; a **gamma camera** (sodium iodide crystal, photomultipliers, collimator) images the photons. SPECT (single-photon emission computed tomography) gives 3D images.
- **^{99m}Tc (technetium-99m):** 6-hour half-life, single 140 keV photon, so it is the workhorse.

Renal tracers

Tracer	Handling	Study	Main uses
DTPA (diethylenetriaminepentaacetic acid)	Glomerular filtration only	Dynamic	GFR (glomerular filtration rate); captopril renography; poor images in renal impairment
MAG3 (mercaptoacetyltriglycine)	Tubular secretion (protein-bound), high extraction	Dynamic	**Split function and obstruction (first choice)**; transplant
DMSA (dimercaptosuccinic acid)	Filtered, then retained by proximal tubules	Static at 2 h	**Pyelonephritis and scars**

Diuretic renography

- **Preparation:** good hydration; catheter if outlet obstruction or neurogenic bladder; cap a nephrostomy.
- **Dose:** 5–10 mCi (185–370 MBq) intravenously.
- **Phases:** perfusion (first minute); function: **split function** from the area under each curve at 1–3 min (normal 45:55 to 55:45); drainage.
- **Furosemide** 0.5 mg/kg (adult maximum 40 mg). Timing: F+20 (classic), F+15, F0 or F−15. Always note when it was given.

Post-furosemide T½ (half-clearance time)	Meaning
<10 min	Not obstructed
10–20 min	Indeterminate
>20 min	Obstructed

- **Curve types (O'Reilly):** I normal; II obstructed (rising); IIIa dilated, not obstructed (falls after furosemide); IIIb equivocal.
- **Pitfalls:** poor function, dehydration, full bladder, capacious pelvis.

Other uses

- **Captopril renography (DTPA):** captopril removes angiotensin II support of GFR in the stenotic kidney. Improvement of ≥10% off ACE (angiotensin-converting enzyme) inhibitors means renovascular hypertension is likely.
- **Transplant:** acute rejection (perfusion falls, worsens) versus delayed graft function (perfusion kept, improves); urine leak.
- **DMSA:** acute pyelonephritis defects resolve; scars persist at about 6 months.
- **Nuclear cystography** for follow-up of reflux (more sensitive, lower radiation than VCUG [voiding cystourethrogram]).

- **Also know:** bone scan (MDP, methylene diphosphonate); labelled leucocyte scans; PSMA (prostate-specific membrane antigen) PET (positron emission tomography). Radiation dose is low.

Diagram to draw: renogram curves: normal, obstructed (rising) and dilated non-obstructed (falls after furosemide).

> **Key points to remember:**
> - DTPA = filtration; MAG3 = secretion (best for obstruction); DMSA = cortex.
> - T½ <10 min not obstructed; >20 min obstructed.
> - Normal split 45–55%; furosemide 0.5 mg/kg.

Rapid Revision Sheet – Paper II

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

II-1. Discuss principles of hemodialysis and vasular access in details. (LAQ)

- Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
- Kt/V minimum 1.2 (target 1.4); URR ≥65%.
- AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s.
- Thrombosis usually sits on a stenosis; steal is treated by banding or DRIL.

II-2. Discuss evaluation for urolithiasis and management of bilateral renal staghorn calculi (LAQ)

- NCCT gold standard (97%/95%); 24-h urine for high-risk formers.
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

II-3a. LASERS in BPH. (SAQ)

- Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- HoLEP is size-independent and superior to TURP.
- GOLIATH: PVP non-inferior to TURP; best for anticoagulated men.

II-3b. Interstitial cystitis (Bladder pain syndrome). (SAQ)

- AUA definition: bladder pain with LUTS >6 weeks, no other cause.
- Hunner lesion: cystoscopy over 50; fulguration or triamcinolone.
- Amitriptyline, PPS, DMSO; potassium test obsolete.

II-3c. Silodosin. (SAQ)

- 8 mg with a meal; 4 mg if CrCl 30–50; avoid if <30.
- Equal to tamsulosin 0.4 mg; greatest BOOI fall.
- Anejaculation is the signature effect; least hypotension.

II-3d. ADPKD (Adult polycystic kidney disease) (SAQ)

- PKD1: chromosome 16, 85%, earlier ESRD.
- Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
- Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

II-3e. RIRS (Retrograde intrarenal surgery) (SAQ)

- 3.6 Fr channel; ≥270° deflection.
- Access sheath lowers pressure but injures the ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.

II-3f. Renal isotope scans (SAQ)

- DTPA = filtration; MAG3 = secretion (best for obstruction); DMSA = cortex.
- T½ <10 min not obstructed; >20 min obstructed.
- Normal split 45–55%; furosemide 0.5 mg/kg.